Selection on Observables II: Estimation Approaches

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Chad Hazlett

UCLA

chazlett@ucla.edu

SOO: Identification and Estimation

Recall the SOO identification assumption:

Assumption: Conditional Ignorability

$$\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp D_i \mid X_i = x \quad \text{for any} \quad x \in \mathcal{X}$$

(a.k.a. exogeneity, unconfoundedness, selection on observables, no omitted variables)

How to actually estimate something that conditions on X?

A direct approach would be:

- estimate effects $\tau(X)$ for each stratum or level of X
- average these, weighting over some distribution, e.g. $p(X)$ or $p(X|D = 1)$.

Some methods do this quite literally:

- sub-classification
- matching

Matching is Not an Identification Strategy

ATT Estimation by Matching

- For each treated unit i with covariates X_i, you would like to estimate $\tau_i = Y_{1i} - Y_{0i}$.
- For treated units you observe Y_{1i}, but where to get Y_{0i}?
- Matching: borrow it from control unit with (nearly) the same X_i
- So estimator is:

$$\widehat{\tau}_{ATT} = \frac{1}{N_1} \sum_{D_i=1} \left(Y_i - Y_{j(i)} \right)$$

where $Y_{j(i)}$ is the outcome of an untreated observation such that $X_{j(i)}$ is the closest value to X_i among the untreated observations.

We can also use the average of M closest matches:

$$\widehat{\tau}_{ATT} = \frac{1}{N_1} \sum_{D_i=1} \left\{ Y_i - \left(\frac{1}{M} \sum_{m=1}^{M} Y_{j_m(i)}, \right) \right\}$$

Does SOO assumption guarantee that this gets you the ATT?

Matching: Example with a Single X

unit	Potential Outcome under Treatment	Potential Outcome under Control		
i	Y_{1i}	Y_{0i}	D_i	X_i
1	6	?	1	3
2	1	?	1	1
3	0	?	1	10
4		0	0	2
5		9	0	3
6		1	0	-2
7		1	0	-4

What is $\widehat{\tau}_{ATT} = \frac{1}{N_1} \sum_{D_i=1} \left(Y_i - Y_{j(i)}\right)$?

Matching: Example with a Single X

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What is $\widehat{\tau}_{ATT} = \frac{1}{N_1} \sum_{D_i=1} \left(Y_i - Y_{j(i)}\right)$?

$\widehat{\tau}_{ATT} = 1/3 \cdot (6 - 9) + 1/3 \cdot (1 - 0) + 1/3 \cdot (0 - 9) = -3.7$

Common Distance Metrics

What do we mean by "closeness" in X when it is multi-dimensional?

Covariate vectors for i, j: $X_i = [X_i^{(1)}, X_i^{(2)}, ..., X_i^{(k)}]^\top$ and $X_j = [X_j^{(1)}, X_j^{(2)}, ..., X_j^{(k)}]^\top$

Some common options:

1. Euclidean distance:

$$||X_i - X_j|| = \sqrt{(X_i - X_j)^\top (X_i - X_j)}$$
$$= [(X_i^{(1)} - X_j^{(1)})^2 + ... + (X_i^{(P)} - X_j^{(P)})^2]^{\frac{1}{2}}$$

2. "StataD" (`nnmatch`)/default in `Match()` for `R`: rescaled Euclidean

$$StataD(X_i, X_j) = \sqrt{(X_i - X_j)^\top diag(\Sigma_X^{-1})(X_i - X_j)}$$

where Σ is the Variance-Covariance-Matrix. Invariant to rescaling of X

Common Distance Metrics

3. Mahalanobis distance:

$$MD(X_i, X_j) = \sqrt{(X_i - X_j)^\top \Sigma^{-1} (X_i - X_j)}$$

where Σ is the Variance-Covariance-Matrix. Invariant to rescaling and rotations of X

4. GeneticD (`GenMatch`):

$$GeneticD(X_i, X_j) = \sqrt{(X_i - X_j)^\top (S^{-1/2})^\top \, W \, S^{-1/2} (X_i - X_j)}$$

where W is a $(P \times P)$ positive definite weight matrix with zeros in off-diagonals, controlling "variable importance"

Mahalanobis Distance: Example

	X_1	X_2
Treated	0	0
Control A	2	2
Control B	1.8	0

$$X_T = \begin{pmatrix} 0 & 0 \end{pmatrix}^\top \quad X_A = \begin{pmatrix} 2 & 2 \end{pmatrix}^\top \quad X_B = \begin{pmatrix} 1.8 & 0 \end{pmatrix}^\top$$

$$\Sigma = \begin{pmatrix} & X^{(1)} & X^{(2)} \\ X^{(1)} & 1 & .9 \\ X^{(2)} & .9 & 1 \end{pmatrix}$$

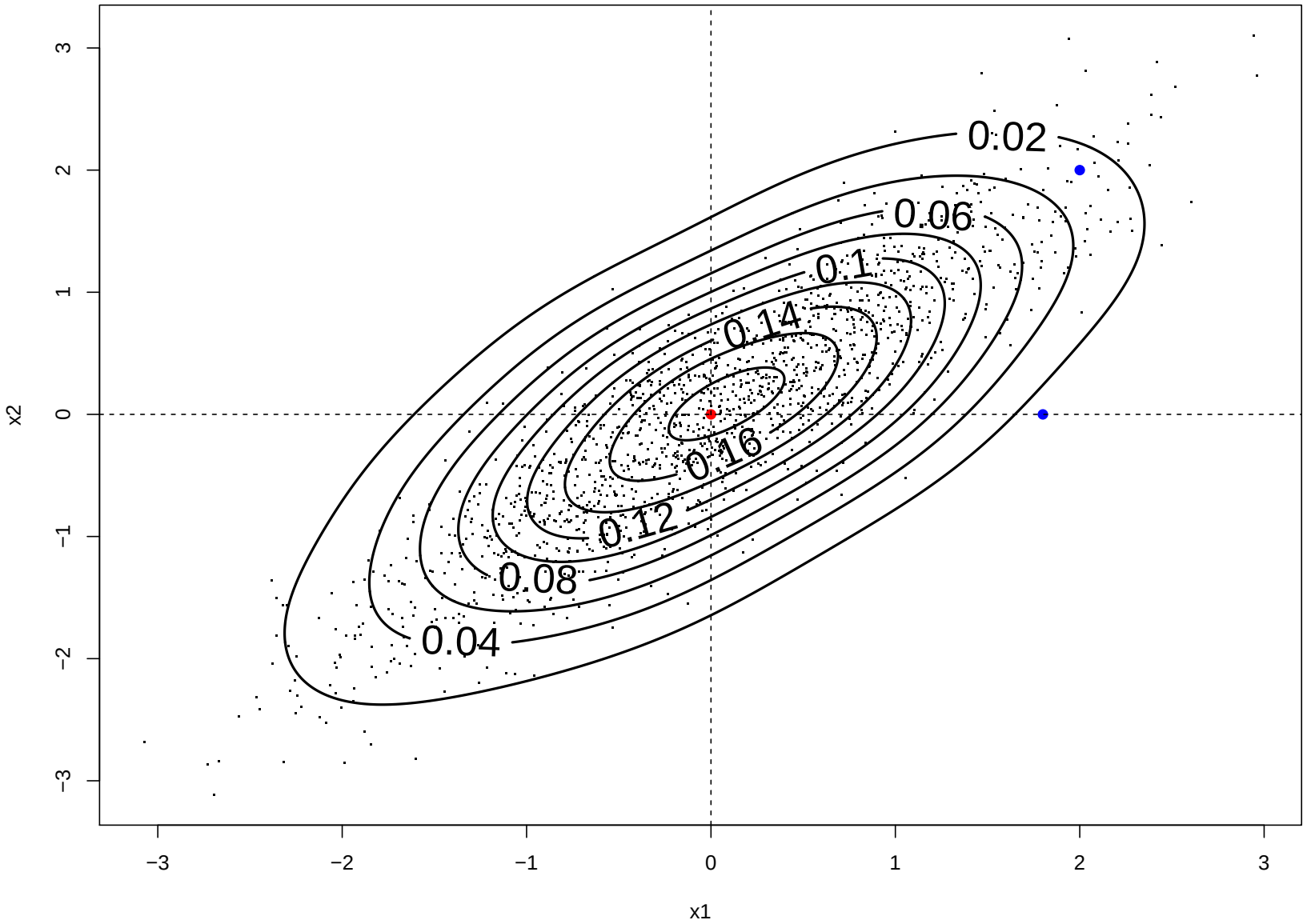
Which control is closer?

Mahalanobis Distance

$$X_T = \begin{pmatrix} 0 & 0 \end{pmatrix}^\top, \; X_A = \begin{pmatrix} 2 & 2 \end{pmatrix}^\top, \; X_B = \begin{pmatrix} 1.8 & 0 \end{pmatrix}^\top, \; \Sigma = \begin{pmatrix} & X^{(1)} & X^{(2)} \\ X^{(1)} & 1 & .9 \\ X^{(2)} & .9 & 1 \end{pmatrix}$$

$$\begin{aligned}
MD(X_A, X_T) &= \sqrt{(X_A - X_T)^\top \Sigma^{-1} (X_A - X_T)} \\
&= \sqrt{[\begin{pmatrix} 2 & 2 \end{pmatrix} - \begin{pmatrix} 0 & 0 \end{pmatrix}] \begin{pmatrix} 1 & .9 \\ .9 & 1 \end{pmatrix}^{-1} [\begin{pmatrix} 2 & 2 \end{pmatrix} - \begin{pmatrix} 0 & 0 \end{pmatrix}]^\top} \\
&= \sqrt{\begin{bmatrix} 2 & 2 \end{bmatrix} \begin{pmatrix} 5.2 & -4.7 \\ -4.7 & 5.2 \end{pmatrix} \begin{bmatrix} 2 & 2 \end{bmatrix}^\top} \\
&= 4.2 \\
MD(X_B, X_T) &= \sqrt{\begin{bmatrix} 1.8 & 0 \end{bmatrix} \begin{pmatrix} 5.2 & -4.7 \\ -4.7 & 5.2 \end{pmatrix} \begin{bmatrix} 1.8 & 0 \end{bmatrix}^\top} \\
&= 17
\end{aligned}$$

Mahalanobis Distance



Local Methods and the Curse of Dimensionality

Problem: the volume increases exponentially in dim(X)

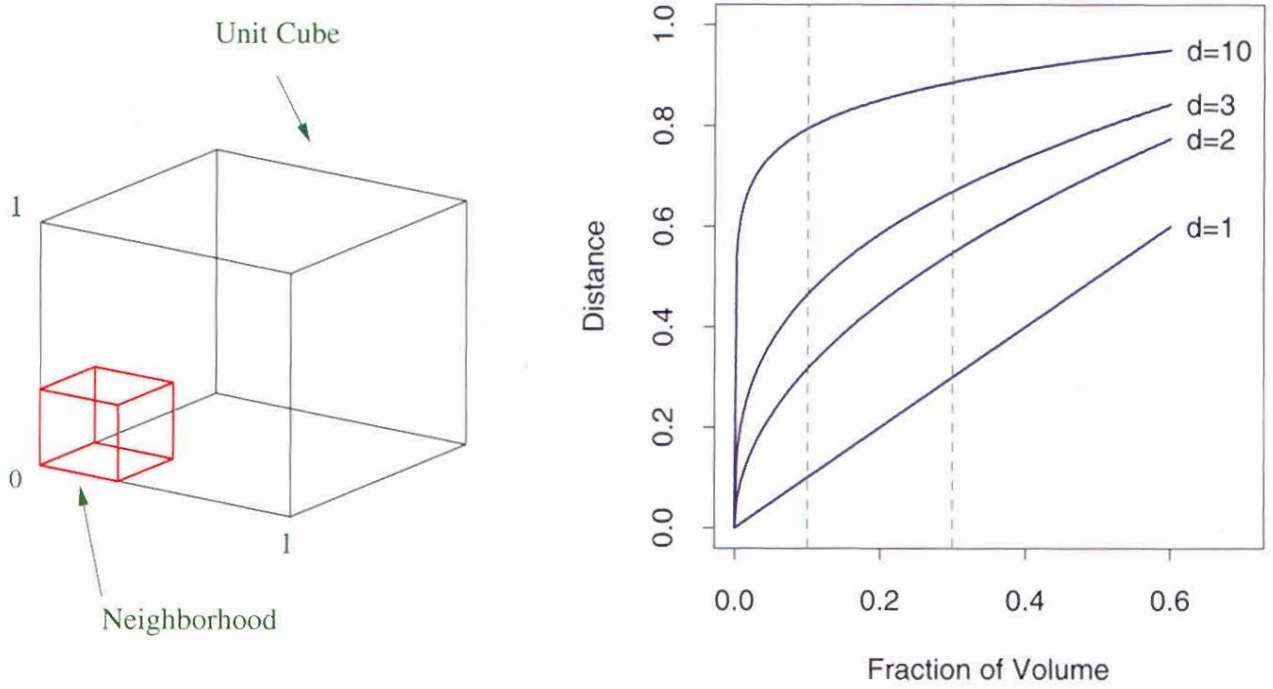


FIGURE 2.6. *The curse of dimensionality is well illustrated by a subcubical neighborhood for uniform data in a unit cube. The figure on the right shows the side-length of the subcube needed to capture a fraction r of the volume of the data, for different dimensions p. In ten dimensions we need to cover 80% of the range of each coordinate to capture 10% of the data.*

Matching with Bias Correction

Weakness of matching: discrepancy=$||X_i - X_{j(1)}||_d > 0$

With multiple continuous variables, this becomes problematic

- if X predicts treatment, then this discrepancy is not just random noise
- does not average away quickly: not $\sqrt{N}$ consistent (Abadie and Imbens, 2005)

Compromise: adjust for discrepancy:

- On average, Y_{0j} differs from Y_{0i} by $\mathbb{E}[Y_0|X_i] - \mathbb{E}[Y_0|X_{j(i)}]$
- Consider population regression function $\mu_0(X) = \mathbb{E}[Y_0|X]$
- Let's estimate $\hat{\mu}_0$ by OLS regression of Y on X among controls

$$\hat{\mu}_0(X) = \beta_0 + \beta_1 X$$

Matching with Bias Correction

So subtract off estimated discrepancy on Y_0 from each pair (Abadie & Imbens, 2005)

$$\tilde{\tau}_{ATT} = \frac{1}{N_1} \sum_{D_i=1} ((Y_i - Y_{j(i)}) - (\widehat{\mu}_0(X_i) - \widehat{\mu}_0(X_{j(i)})),$$

- these "bias-corrected" matching estimators are an improvement even if $\hat{\mu_0}$ is misspecified
- the large sample distribution of this estimator (for the case of matching with replacement) is roughly normal.

In R: `Match(Y,Tr, X,BiasAdjust = TRUE)`

Bias Adjustment with Matched Data

unit	Potential Outcome under Treatment	Potential Outcome under Control		
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5		9	0	3
6		1	0	8

What is $\tilde{\tau}_{ATT} = \frac{1}{N_1} \sum_{D_i=1} \left((Y_i - Y_{j(i)}) - (\widehat{\mu}_0(X_i) - \widehat{\mu}_0(X_{j(i)})) \right)$?

Bias Adjustment with Matched Data

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4		0	0	2
5		9	0	3
6		1	0	8

What is $\tilde{\tau}_{ATT} = \frac{1}{N_1}\sum_{D_i=1}\Big((Y_i - Y_{j(i)}) - (\widehat{\mu}_0(X_i) - \widehat{\mu}_0(X_{j(i)}))\Big)$?

Estimate $\widehat{\mu}_0(x) = \beta_0 + \beta_1 x = 5 - .4x$. Now plug in:

$$\begin{aligned}
\widehat{\tau}_{ATT} &= 1/3\{((6-9) - (\widehat{\mu}_0(3) - \widehat{\mu}_0(3))) \\
&+ ((1-0) - (\widehat{\mu}_0(1) - \widehat{\mu}_0(2))) \\
&+ ((0-1) - (\widehat{\mu}_0(10) - \widehat{\mu}_0(8)))\} \\
&= \frac{1}{3}[(6-9) + [(1-0) - (4.6-4.2)] + [(0-1) - (1-1.8)]] \\
&= -0.87 \ \text{(rather than -1)}
\end{aligned}$$

Checking balance: Theory

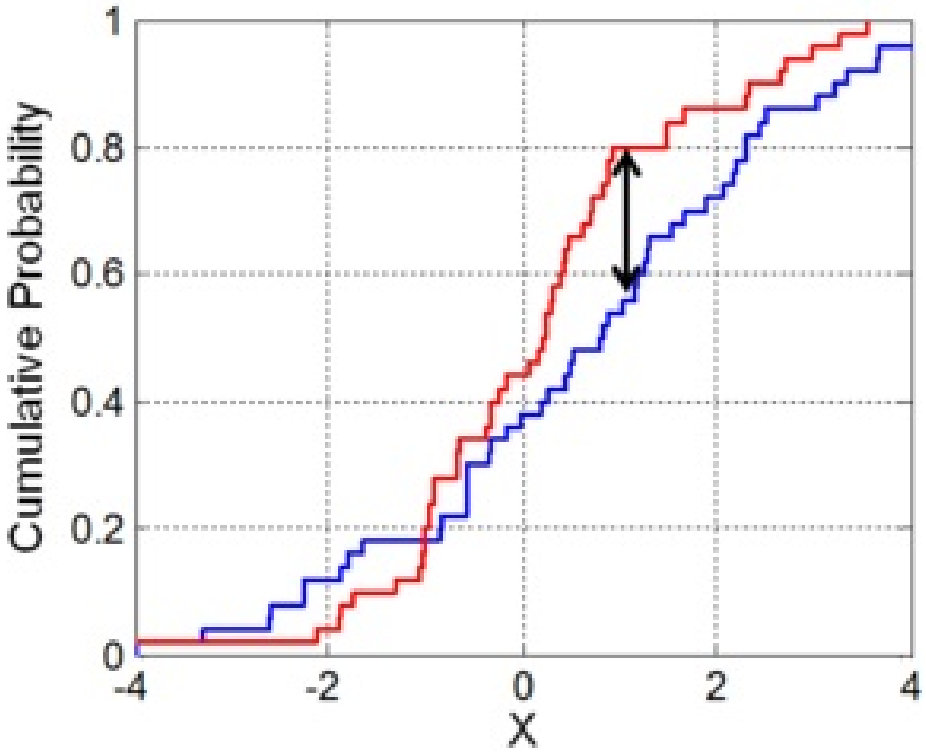
- Ideally, compare the joint distribution of all X_i between the treated and untreated in the matched sample
- In practice, compare various low-dimensional summaries of $F(x)$ (mean difference to compare expectations, variance ratio to compare variances, etc.)
- Balance tests are often used (e.g. t-test, F-test, KS test)
- Note that hypothesis tests can be misleading in a matching context: you can make everything insignificant by simply dropping lots of observations — make sure this is not happening (see Hartman & Hidalgo)

One common workflow:

- Estimate $\rightarrow$ Check Balance $\rightarrow$ Re-estimate $\rightarrow$ Check Balance $\rightarrow \cdots$ (ad infinitum until you get a good balance)
- Is this data snooping? No, because blind to Y

Kolmogorov-Smirnov (KS) Test

- The KS test is used to test whether two random variables are sampled from the same distribution
- The test is nonparametric, meaning that it works (asymptotically) without assumptions about the form of the underlying distribution
- Consider N observations of two random variables, X_0 and X_1.



Checking balance: Theory

The (two-sample) KS statistic:

$$D = \sup_x \left| \widehat{F}_1(x) - \widehat{F}_0(x) \right|$$

where $\widehat{F}_0(x)$, $\widehat{F}_1(x)$ is the empirical CDF of X_0, X_1.

The KS null hypothesis: $F_1(x) = F_0(x)$ (no difference in true distributions)

Under the null, D has the Kolmogorov distribution as $n \to \infty$.

Reject the null at level α if

$$\frac{D}{\sqrt{(n_1 + n_0)/n_1 n_0}} > c_\alpha$$

level (α)	.1	.05	.01
critical value (c_α)	1.22	1.36	1.63

Balance Checks

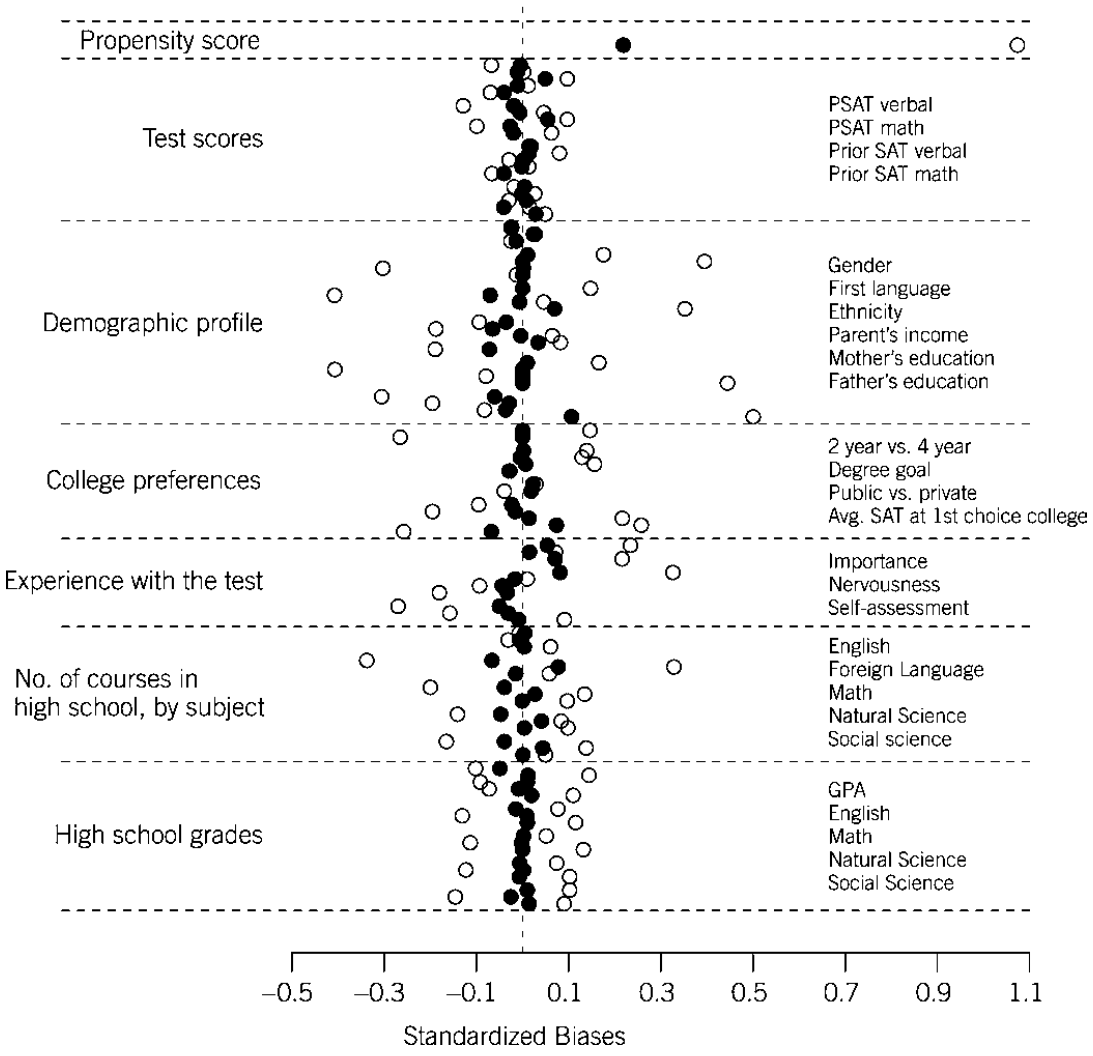


Figure 3. Standardized Biases Without Stratification or Matching, Open Circles, and Under the Optimal [.5, 2] Full Match, Shaded Circles.

Balance Checks

TABLE 2. Balance Summary Statistics and Tests: Russian and Chechen Sweeps

Pretreatment Covariates	Mean Treated	Mean Control	Mean Difference	Std. Bias	Rank Sum Test	K-S Test
Demographics						
Population	8.657	8.606	0.049	0.033	0.708	0.454
Tariqa	0.076	0.048	0.028	0.104	0.331	—
Poverty	1.917	1.931	−0.016	−0.024	0.792	1.000
Spatial						
Elevation	5.078	5.233	−0.155	−0.135	0.140	0.228
Isolation	1.007	1.070	−0.063	−0.096	0.343	0.851
Groznyy	0.131	0.138	−0.007	−0.018	0.864	—
War Dynamics						
TAC	0.241	0.282	−0.041	−0.095	0.424	—
Garrison	0.379	0.414	−0.035	−0.072	0.549	—
Rebel	0.510	0.441	0.070	0.139	0.240	—
Selection						
Presweep violence	3.083	3.117	−0.034	0.009	0.454	0.292
Large-scale theft	0.034	0.055	−0.021	−0.115	0.395	—
Killing	0.117	0.090	0.027	0.084	0.443	—
Violence Inflicted						
Total abuse	0.970	0.833	0.137	0.124	0.131	0.454
Prior sweeps	1.729	1.812	−0.090	−0.089	0.394	0.367
Other						
Month	7.428	6.986	0.442	0.130	0.260	0.292
Year	2004.159	2004.110	0.049	0.043	0.889	1.000

Note: 145 matched pairs. Matching with replacement.

Choices when Matching

- With or without replacement?
- How many matches?
- Which matching algorithm?
  - genetic matching, kernel matching, full matching
  - coarsened exact matching
  - matching as pre-processing
  - propensity score matching
- General rule: use whatever gives you the best balance subject to conditioning on what you need and not dropping too many units.
- Adjustment methods help, but rely on a model to impute missing potential outcomes.

Variance of the Matching Estimator

Option 1. Ignore the matching uncertainty and estimate the SEs from whatever model you run on the matched sample

- treats the matched sample as fixed
- thus ignores uncertainty in matching

Option 2. Bootstrap

- NO. Not consistent
- "extreme non-smoothness" (Abadie & Imbens 2006)

Option 3. Abadie-Imbens asymptotic SEs

- uses matched pairs to estimate local variance in Y_0 and takes weighted sum that accounts for matching
- provided by `Match()` package.
- generally use these
- but still an open area of research

Useful Matching Functions

The workhorse model is the `Match()` function in the `Matching` package:

```
Match(Y = NULL, Tr, X, Z = X, V = rep(1, length(Y)),
    estimand = "ATT", M = 1, BiasAdjust = FALSE, exact = NULL,
    caliper = NULL,  replace = TRUE, ties = TRUE,
    CommonSupport = FALSE, Weight = 1, Weight.matrix = NULL,
    weights = NULL, Var.calc = 0,  sample = FALSE, restrict = NULL
    match.out = NULL, distance.tolerance = 1e-05,
    tolerance = sqrt(.Machine$double.eps), version = "standard")
```

Default distance metric is normalized Euclidean distance

- `MatchBalance` for balance checking
- `GenMatch` for genetic matching

Example: Blattman and Annan (2009)

The Consequences of Child Soldiering. The Review of Economics and Statistics.

Question: what is the impact of abduction by the rebel group *Lord's Resistance Army* on education?

ID Strategy: SOO. According to the logic by which abduction occurred, we will assume that abduction is indiscriminate, conditional on age and location.

> *Abduction was large-scale and seemingly indiscriminate; 60,000 to 80,000 youth are estimated to have been abducted and more than a quarter of males currently aged 14 to 30 in our study region were abducted for at least two weeks...*
>
> *Youth were typically taken by roving groups of 10 to 20 rebels during night raids on rural homes. Adolescent males appear to have been the most pliable, reliable and effective forced recruits, and so were disproportionately targeted by the LRA. Youth under age 11 and over 24 tended to be avoided and had a high probability of immediate release.*

Example: Blattman and Annan (2009)

Data: panel survey of male youth in war-afflicted regions of Uganda.

- `abd`: abducted by the LRA (the treatment)
- `c_ach - c_pal`: Location indicators
- `age`: age in years
- `fthr_ed`,`mthr_ed` : father's/mother's education (years)
- `orphan96`: indicator if parent's died before 1997
- `hh_fthr_frm`: indicator if father is a farmer
- `hh_size96`: household size in 1996
- `educ`: years of education
- `distress`: index of emotional distress (0-15)
- `logwage`: log of average daily wage earned in last 4 weeks

Which of these **must** you match on given the ID strategy?

After matching, which of these should you expect good balance on once NB: `educ`, `distress`, and `logwage` are measured after abduction.

Code example(s)

Check online for example code of matching and balance testing, using both older and newer packages.